

0100CXT308062401

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree (R, S) Examination May 2024 (2019 Scheme)

Course Code: CXT308**Course name: COMPREHENSIVE COURSE WORK**

Max. Marks: 50

Duration: 1Hour

- Instructions:**
- (1) Each question carries one mark. No negative marks for wrong answers
 - (2) Total number of questions: 50
 - (3) All questions are to be answered. Each question will be followed by 4 possible answers of which only ONE is correct.
 - (4) If more than one option is chosen, it will not be considered for valuation.

1. Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function $x \bmod 10$, which of the following statements are true?

i) 9679, 1989, 4199 hash to the same value

ii) 1471, 6171 has to the same value

iii) All elements hash to the same value

iv) Each element hashes to a different value

a) i only

b) ii only

c) i and ii

d) iii or iv
2. The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)

a) 2

b) 3

c) 4

d) 5
3. The average depth of binary tree with N elements

a) $O(N)$

b) $O(N^2)$

c) $O(\log N)$

d) $O(\sqrt{N})$
4. Which data structure is needed to convert infix notation to postfix notation

a) queue

b) stack

c) tree

d) graph
5. Which of the following is NOT a common operation in a queue data structure?

a) Enqueue

b) Dequeue

c) Peek

d) Shuffle
6. Which data structure is commonly used to implement the event-driven simulation of complex systems, such as in computer network simulations or traffic simulations?

a) queue

b) stack

c) tree

d) array
7. Which of the following algorithms can be used to most efficiently determine the presence of a cycle in a given graph

a) DFS

b) BFS

c) Kruskals algorithm

d) Prims algorithm

8. Depth First Search is equivalent to which of the traversal in the Binary Trees?
a) Preorder b) Inorder c) Postorder d) Level order
9. Which is the most appropriate data structure for reversing a word?
a) queue b) linked list c) tree d) stack
10. The process which is copied from main memory to secondary memory on the basis of requirement is known as
a) segmentation b) paging c) demand paging d) threads
11. Which of the following is major part of time taken when accessing data on the disk?
a) rotational latency b) waiting time c) seek time d) response time
12. FIFO cpu scheduling is a type of ----- scheduling
a) priority b) pre-emptive c) deadline d) non-premptive
13. How many minimum variables is/are required to share between processes, so as to solve the critical section problem?
a) 1 b) 2 c) 3 d) 0
14. Out of these page replacement algorithms, which one suffers from Belady's anomaly?
a) LRU b) LFU c) FIFO d) Optimal
15. In every entry of a page table, the essential content(s) is/are:
a) page frame number b) virtual page number c) both a and b d) accessing right information
16. Which of the following is also known as the base register?
a) Relocation register b) Program counter c) Instruction register d) Limit register
17. All unsafe states are
a) deadlock states b) not deadlock states c) none d) secure states
18. A counting semaphore S is initialized to 10. Then, 6 P operations and 4 V operations are performed on S. What is the final value of S?
a) 10 b) 8 c) 2 d) 6
19. Memory fragmentation results in_____
a) stack overflow b) poor usage of memory c) page fault d) better memory allocation
20. The segment base contains the
a) Starting logical address of process b) Starting physical address of the segment in memory c) Length of segment d) None of these
21. Which of the following is a type of computer architecture used in now a days

- 22 a) Harvard Architecture b) Microarchitecture c) Von-Neumann Architecture d) System Design
Control signals are generated in different types of system architectures.
- 23 a) Hardwired b) Microprogrammed c) Both a and b d) Micro-instruction
In the following indexed addressing mode instruction, MOV 5(R1), LOC the effective address is _____
- 24 a) EA = 5+R1 b) EA = R1 c) EA = [R1] d) EA=5+[R1]
The performance of a computer system is primarily influenced by:
- 25 a) Clock speed of CPU b) Size of main memory c) I/O bandwidth d) All the above
Booth's Algorithm is primarily used for
- 26 a) Signed binary multiplication b) Unsigned binary multiplication c) Signed binary addition d) Floating-point number multiplication
In Direct Memory Access (DMA), which device gains control of the system bus?
- 27 a) CPU b) DMA Controller c) Both CPU and DMA controller can access the bus at the same time d) CPU and DMA controller take turns accessing the bus
In associative mapping, a memory block can be placed in:
- 28 a) A specific cache line based on its address b) A set of pre-defined cache lines c) Any available cache line d) none of these
The correct sequence of HTML tags for starting a webpage is -
- 29 a) Head, Title, HTML, body b) HTML, Body, Title, Head c) HTML, Head, Body, Title d) HTML, Head, Title, Body
Which of the following tag is used for inserting the largest heading in HTML?
- 30 a) <h1> b) <h6> c) <h3> d) <h4>
How to create an unordered list (a list with the list items in bullets) in HTML?

- a) b) <i> c) d)
- 31 Which of the following is used to read an HTML page and render it?
- a) Web server b) Web browser c) Operating system d) Web network system
- 32 Which of the following tags are mandatory in an HTML document?
- a) <head> and <body> b) <title> and <body> c) <HTML> and <head> d) <HTML> only
- 33 Which of the following elements are considered 'block-level' elements in HTML?
- a) and <a> b) <p> and <div> c) and <i> d) and
- 34 Which of the following keywords is used to define a variable in Javascript?
- a) var b) let c) none of these d) both a and b
- 35 Which of the following methods can be used to display data in some form using Javascript?
- a) document.write() b) console.log() c) window.alert() d) all of these
- 36 HMD stands for?
- a) Head masked display b) Head mounted display c) Head made display d) Head mounted detection
- 37 VR headset is a
- a) a type of glove b) device worn on feet c) a specialised chair d) head mounted display that covers eyes and ears
- 38 Which technology enables users to walk and move naturally in a VR environment?
- a) Hoverboard b) Treadmill c) Unicycle d) Segway
- 39 Which type of VR is designed to be experienced with a web browser without the need for a dedicated VR headset?
- a) Augmented Reality b) Virtual Reality(VR) c) Mixed Reality d) WebVR
- 40 Which of the following is NOT a common challenge in VR development?
- a) Motion sickness b) Latency c) High resolution displays d) Hardware compatibility

- 41 A term for Illusion of immersion by projecting stereo images on the walls and floor of a room.
- a) GUI b) HMD c) BOOM d) CAVE
- 42 What are the advantages of Industry 4.0?
- a) Improved productivity and efficiency b) Low cost of implementation c) Creating more vacancies for workers d) No risk of hacking into the internal network
- 43 What type of CSS is the following code snippet
- ```
<h1 style="color:blue;">A Blue Heading</h1>
```
- a) Inline                      b) Internal                      c) External                      d) none of these
- 44 Which of the following is an example of a loosely typed language?
- a) C                      b) C++                      c) Java                      d) Javascript
- 45 Which of the following is a linear data structure?
- a) AVL tree                      b) Binary Tree                      c) Graph                      d) Array
- 46 Which of the following sorting algorithms provide the best time complexity in the worst-case scenario?
- a) Quick sort                      b) Selection Sort                      c) Merge Sort                      d) Bubble sort
- 47 Which data structure is mainly used for implementing the recursive algorithm?
- a) stack                      queue                      c) array                      d) list
- 48 If the wait for graph contains a cycle indicate
- a) then a deadlock does not exist                      b) then a deadlock exist                      c) then the system is in a safe state                      d) none of theses
- 49 Which of the following is a disk scheduling algorithm
- a) Round robin                      b) Elevator                      c) LRU                      d) Bankers
- 50 The number successful accesses to memory stated as a fraction is called as \_\_\_\_\_
- a) miss rate                      b) hit rate                      c) access rate                      d) success rate